Shubhadeep Mandal

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Education

International Institute of Information Technology, Hyderabad July 2025 – Present
Telangana, India
Master of Technology in Computer Science and Engineering *CGPA: 9.12*

Indian Institute of Information Technology, Ranchi Aug 2020 – Jun 2024
Ranchi, India
Bachelor of Technology (Hons.) in Computer Science and Engineering *CGPA: 9.05*

Research and Open Source

RookDB: Open-Source Relational Database, IIIT Hyderabad | *Rust, SQL, Databases*

- Core contributor and co-designer of the database: storage engine, buffer pool, B+Tree indexes, query planner and executor.
- Built a random-query tester that checks results against SQLite; **50,000 queries** match, and every bug it found is now a test.
- Made schema changes crash-safe, so a crash mid-change leaves either the old catalog or the new one, never a mix.
- Led performance work that made the benchmark suite **14x faster**, including a hash join for tables larger than memory.

Projects

Negotiate, Then Solve: LLM Agents for Timetabling | *Python, Gemini, Qwen, RAG, CP-SAT, FastAPI*

- Built an LLM assistant that turns scheduling requests into constraints for a solver and negotiates conflicts with people.
- Added safety checks (authority, cited rules, human approval); **no attack succeeded** in 50 prompt-injection tests.
- Built hybrid keyword and embedding search over policy documents, reaching **93%** recall.
- Fine-tuned a small 4B model with LoRA that runs on a laptop GPU and outperformed the Gemini model it learned from.

Warehouse Traffic Control: Distributed Robot Coordination | *JavaScript, Three.js, Web Workers*

- Built a simulator where 16 region managers coordinate 200 warehouse robots instead of one central server.
- Designed a lease-based protocol with **zero collisions** over 416K moves, even with lost messages, clock drift and crashes.
- Added deadlock detection and crash recovery, so a failed manager rebuilds its state in seconds.
- Compared with a central coordinator: **7.6x lower** peak load per node, and a crash affects 11 robots instead of 120.

Peer-to-Peer Distributed File Sharing System | *C++, POSIX Sockets, Multithreading, OpenSSL, Linux*

- Built a BitTorrent-style file sharing system in C++ with a central tracker, multithreaded peers and per-piece integrity checks.
- Profiled and fixed bottlenecks in scheduling and networking, cutting a 1GB download from **193 s to under 13 s**.
- Added a replicated backup tracker with automatic failover, tested with unit tests under sanitizers.

Technical Skills

Languages: C/C++, Rust, Python, TypeScript, JavaScript, SQL, Bash.
Systems: Linux, POSIX APIs, Multithreading, Sockets, OpenSSL.
Distributed: gRPC, Protobuf, Hadoop, MPI, SLURM.
Other: PyTorch, React, Next.js, PostgreSQL, Git, Docker.

Achievements

- **B.Tech Rank Holder:** Ranked 2nd by GPA in the B.Tech CSE batch at IIIT Ranchi.
- **PGEE 2025:** Secured All-India Rank 100 in the IIIT Hyderabad Post Graduate Entrance Examination.
- **Megathon 2025:** Placed 3rd in the mobile computing track.

Interests

- Reading web novels and manga.
- Playing chess and badminton.